

**CHIEF GENERAL MANAGER
DERIVATIVE CELL**

SMDRP/DC/CIR- 10/01
November 2, 2001

**To,
The Chief Executive Officer/ Managing Director
of Derivative Segment of NSE & BSE
and their Clearing House / Corporation.**

Dear Sir,

Sub: Scheme for introduction of Single Stock Futures and the Risk Containment Measures.

This is in continuation of SEBI Circular No. IES/DC/CIR-4/99 dated July 28, 1999, Circular No. IES/DC/CIR-5/00 dated December 11, 2000 and Circular No. SMD/DC/Cir-10/01 dated June 20, 2001 wherein the risk containment measures for Exchange traded Index Futures, Index Option, and Stock Option Contracts were laid down.

SEBI has setup an Advisory Committee on Derivatives headed by Prof. J. R Varma. The Advisory Committee had recommended the introduction of Single Stock Futures in the Indian Securities Market. SEBI Board in its meeting on September 4, 2001 considered the recommendation of the Advisory Committee on Derivatives and granted in-principle approval for the introduction of futures on 31 stocks, in which options contracts have been permitted by SEBI. Further, the Board desired that the Advisory Committee on Derivatives,

- devise a detailed scheme for the introduction of this product,
- specify the risk containment measures which could include cash settlement of Single stock futures, and
- specify the time frame in which futures on Single stocks could be introduced after assessing the preparedness of the exchanges in terms of infrastructure.

On the risk containment measures the Advisory Committee on Derivatives agreed to adopt the existing risk management framework in the derivative market for Single Stock Futures. The committee asked SEBI to set various parameters like volatility estimates, price range, minimum margin level, calendar spread charges etc, in consultation with the exchanges.

The scheme of introduction and the risk containment measures for Single stock futures contracts framed in consultation with the stock exchanges, within the framework specified by the Advisory Committee on Derivatives, and approved by the SEBI Board in its meeting On November 1, 2001, are as follows-

1. The Single Stock Futures Contracts to be traded on the derivative exchange/segments shall have prior approval of SEBI. The Contract should comply with the disclosure requirements, if any, laid down by SEBI.
2. The Exchanges shall initially introduce Single Stock Futures Contract, which shall be settled in cash.
3. In order to maintain symmetry between Futures and Options Contract on the same underlying, the lot size of the option contract and the multiplier for the futures contract shall remain the same for the given underlying.
4. Single Stock Futures contract shall be permitted upto maximum maturity of 12 months. Initially, Single Stock Futures contract shall have maturity of three months and at the time of its introduction, three contracts of maturity of one-month, two-month and three-month would be introduced simultaneously. Therefore, at any point in time atleast three Single Stock Futures contracts on a particular underlying would be available for trading.
5. Any adjustments in contract specifications of Single Stock Futures Contract at the time of corporate actions would be in line with international best practices and the sub-committee on adjustment of corporate actions for Stock Options would decide on any such adjustments.
6. **Eligibility of Stocks**

The Board has approved the introduction of Single Stock Futures Contracts on 31 stocks on which option contracts have been introduced on BSE & NSE. The Advisory Committee on Derivatives shall review the eligibility criteria for introduction of futures and options on any other stock from time to time.

7. Risk Containment Measures

A portfolio based margining approach shall be adopted which will take an integrated view of the risk involved in the portfolio of each individual client comprising of his positions in all Derivative Contracts i.e. Index Futures, Index Option, Stock Options and Single Stock Futures. The parameters for such a model should include the following-

A. initial margin or worst scenario loss

The Initial Margin requirements are based on worst scenario loss of a portfolio of an individual client to cover 99% VaR over one day horizon across various scenarios of price changes and volatility shifts. For Index products the price scan range is specified at three standard deviation (3 sigma) and the volatility scan range is specified at 4%. For stock option contracts the price scan range is specified at three and a half standard deviation (3.5 sigma) and the volatility scan range is specified at 10%. There is also a minimum margin requirement. For index futures contracts it is specified that in no case the initial margin shall be less than 5% of the value of the contract. For index options a short option minimum charge of 3% of the notional value of all short index option has been prescribed and in the case of stock option contracts, a short option minimum charge of 7.5% of the notional value of all short stock option contract has been prescribed.

In the case of **Single Stock Futures**, the initial margin would be computed as the worst scenario loss of a portfolio comprising of all the positions of a client in all the futures and options contracts. For Single Stock Futures the price scan range would be 3.5 Standard Deviation (3.5 sigma) and in no case the initial margin for Single Stock Futures contract shall be less than 7.5% of the value of the Single Stock Futures contract. The SPAN margining system, which has been adopted by both BSE & NSE, does not have the provision to provide for charging a minimum margin of 7.5% for futures contracts. However, in order to achieve the requirement of minimum margin for the Single Stock Futures contract, the price scan range would be adjusted so as to ensure that the initial margin for Single Stock Futures contracts does not fall below 7.5% in any scenario. The standard deviation would be calculated as per the methodology specified in the Prof. J. R Varma Committee Report on the Risk Containment Measures for Index Futures.

The Initial Margin requirement shall continue to be netted at level of individual client and shall be calculated on a gross basis at the level of Trading / Clearing Member. The Initial margin requirement for the proprietary position of Trading/Clearing member shall be calculated on a net basis

B. Calendar spread for the Single Stock Futures contract

On the introduction of option contracts, the margin on calendar spread is calculated on the basis of delta of the portfolio consisting of futures and option contract in each month. Thus, a portfolio consisting of a near month option with a delta of 100 and a far month option with a delta of -100 would bear a spread charge equal to the spread charge for a portfolio which is long 100 near month futures and short 100 far month futures. The Calendar Spread Margin is charged in addition to the Worst Scenario Loss of the portfolio. A calendar spread would be treated as a naked position in the far month contract as the near month contract approaches expiry. A calendar spread is treated as a naked position in the far month contract three trading days before

the near month contract expires. **The same provision shall also apply for calendar spreads in Single Stock Futures contracts.**

C. Exposure Limits

It has been prescribed that the notional value of gross open positions at any point in time in the case of Index Futures and all Short Index Option Contracts shall not exceed 33 1/3 (thirty three one by three) times the liquid networth of a member, and in the case of Stock Option Contracts, the notional value of gross short open position at any point in time shall not exceed 20 (twenty) times the liquid networth of a member.

In the case of **Single Stock Futures Contracts** the value of gross open positions at any point in time in all the Single Stock Futures contracts shall not exceed 20 (twenty) times the available liquid networth of a member. Therefore, the exchanges would be required to ensure that 5% of the notional value of gross open position in Single Stock Futures contracts is collected /adjusted from the liquid networth of a member on a real time basis. Exposure limits are in addition to the initial margin requirements.

Exposure limit for calendar spreads in the case of Single Stock Futures contracts: As prescribed in the case of index futures contract, the Calendar Spread shall be regarded as an open position of one third (1/3rd) of the mark to market value of the far month contract. As the near month contract approaches expiry, the spread shall be treated as a naked position in the far month contract three days prior to the expiry of the near month contract. **The same provision shall apply to the spread positions in the case of Single Stock Futures contract.** If the closing out of one leg of a calendar spread causes the members' liquid net worth to fall below the minimum levels specified, his terminal shall be disabled and the clearing corporation / house shall take steps to liquidate sufficient positions to restore the members' liquid net worth to the levels mandated.

D. Real Time Computation

The computation of Worst Scenario Loss has two components. The first is the valuation of the portfolio under sixteen scenarios. At the second stage, these Scenario Contract Values are applied to the actual portfolio positions to compute the portfolio values and the initial margin (Worst Scenario Loss). For computational ease, exchanges are permitted to update the Scenario Contract Values only at discrete time points each day and the latest available Scenario Contract Values would be applied to member/client portfolios on a real time basis.

However, in order to ensure that the most recent scenario are applied for computation of the portfolio values and the initial margin, on introduction of **Single Stock Futures**, it shall be prescribed that the scenario contract values shall be updated atleast 5 times in the day, which may be carried out by taking the closing price of the previous day at the start of trading and the prices at 11:00 a.m., 12:30 p.m., 2:00 p.m., and at the end of the trading session.

E. Margin Collection and Enforcement

As prescribed in the case of index futures contract, the mark to market settlement of Single Stock Futures contracts shall also be collected before start of the next day's trading, in cash. If mark to market margins are not collected before start of the next day's trading, the clearing corporation/house shall collect correspondingly higher initial margin to cover the potential for losses over the time elapsed in the collection of margins. The higher initial margin shall be calculated in the same manner as specified in the Prof. J.R Varma committee reports on risk containment measures for index futures.

The daily closing price of Single Stock Futures Contract for Mark to Market settlement would be

calculated on the basis of the last half an hour weighted average price of the contract. In the absence of trading in the last half an hour the theoretical price would be taken. The Derivative Exchanges/ Segment shall define the methodology of calculating the 'theoretical price' at the time of making an application for approval of the stock futures contract to SEBI and methodology for calculating the 'theoretical price' would also be disclosed to the market. In addition, the exchange shall also specify the methodology for arriving at the closing price at the time of expiry.

The initial margin (or the worst scenario loss) plus the calendar spread charge shall be adjusted against the available Liquid Networth of the member. The members in turn shall collect the initial margin from their clients.

F. Position Limits

On the introduction of index futures contracts, index options contracts and stock options contracts the trading member level and the market wide position limits were prescribed. However, with the introduction of Single Stock Futures contracts, a customer level position limit is also prescribed to deter and detect concentration of positions and market manipulation. The market wide position in the case of stock specific derivative contract (both stock options and Single Stock Future) shall be applicable on the cumulative open positions in derivative contracts on that stock at an Exchange. The volumes in the derivative markets are growing steadily and therefore, position limits shall be reviewed by the Advisory Committee on Derivatives from time to time and also the Advisory Committee shall be empowered to weed out any operational issue in implementation of the position limits.

Client / Customer level position limits:

The gross open position across all derivative contracts on a particular underlying of a customer/client should not exceed the **higher of** –

- 1% of the free float market capitalisation (in terms of number of shares).

or

- 5% of the open interest in the derivative contracts on a particular underlying stock (in terms of number of contracts).

This position limits would be applicable on the combine position in all derivative contracts on an underlying stock at an exchange.

At present the trading system of the exchange requires that client ID should be provided for each trade. However, this client ID assigned by the trading member is not unique to a client across the market. At present the exchange monitors the trading member level position limits however, the client wise limit is not monitored by the exchange and is a requirement of disclosure by the client to the trading member and to the Exchange. With the introduction of Single Stock Futures Contracts the exchanges shall develop a system to monitor client level position limits through their computer system not only with respect to one trading member but also across all the trading members through whom the client is trading in the derivative markets. This would be in addition to the disclosure requirement prescribed for clients. This would require setting up of a system and database which would capture all the details of clients of all the trading members. The system then would identify a common client, who may be trading through more than one trading member, on the basis of some key fields. The setting up of such a database and the system would be required to be completed in a time bound manner in various stages as under: -

- In the first stage, the exchange shall immediately commence monitoring client level position using the client ID given by the trading member.
- In the second stage, the exchange shall collect a unique client identification key which could be PAN number, Depository number etc. and on the basis of that key, the position of the client across all trading members in an exchange could be monitored at least once in the life time of a contract. This could take three to four months for the exchanges to implement.

- In the final stage, exchanges shall collect complete particulars of all the clients from the trading members. The exchange shall develop a system to identify a client trading through more than one trading member and monitor his cumulative positions taken through all the members of the exchange. This would take a longer time and would be implemented simultaneously for both the cash segment and derivative segment, as the database shall be common for both the markets.

Trading Member Level:

At the trading member level the position limit in derivative contracts on a particular stock would be at 7.5% of the open interest or Rs 50 crore whichever is higher for derivative contract in a particular underlying at an exchange. The exchanges shall however specify lower Trading Member level limits for generating surveillance alerts.

Once a member reaches the position limit in a particular underlying then the member shall be permitted to take only offsetting positions (which result in lowering the open position of the member) in derivative contracts on that underlying. In the event, that the position limit is breached due to the reduction in the overall open interest in the market, the member shall be permitted to take only offsetting positions (which result in lowering the open position of the member) in derivative contract in that underlying and no fresh positions shall be permitted. The position limit at trading member level shall be computed on a gross basis across all clients of the Trading member.

Market wide limits:

The market wide limit of open positions (in terms of the number of underlying stock) on an **option and futures contract on a particular underlying stock would be lower of –**

- 30 times the average number of shares traded daily, during the previous calendar month, in the cash segment of the Exchange,

or

- 10% of the number of shares held by non-promoters i.e. 10% of the free float, in terms of number of shares of a company.

This market wide limit prescribed for stock option contracts has been increased from 20 times the average number of shares traded daily to 30 times as these limits would be applicable for both futures and options contracts on a particular underlying stock. Therefore, all the open position in all futures and option contracts on a particular underlying should not exceed the aforementioned market wide position limit.

When the total open interest in a contract reaches 80% of the market wide limit in that contract, the exchanges would double the price scan range and volatility scan range specified. The exchanges are required to continuously review the impact of this measure and take further proactive risk containment measures as may be appropriate, including, further increases in the scan ranges and levying additional margins. Additionally, the exchanges may also set alerts at lower levels for internal surveillance.

1. The Derivative Exchange/Segment shall submit their proposal for approval of the Single Stock Futures Contracts to SEBI which shall include:
 - a. the details of proposed derivative contract to be traded on the exchange which would include:
 - i. Symbol
 - ii. Underlying
 - iii. Multiplier
 - iv. Last Trading Day
 - v. Margins
 - vi. Methodology for calculating closing price for mark to market settlement.
 - vii. Methodology for calculating closing price at time of expiry
 - viii. Trading Hours

- a. the economic purpose it is intended to serve,
- b. likely contribution to market development,
- c. the safeguards and the risk protection mechanism adopted by the exchange to ensure market integrity, protection of investors and smooth and orderly trading,
- d. the infrastructure of the exchange and the surveillance system to effectively monitor trading in Single Stock Futures contracts,
- e. details of settlement procedures & systems with regard to Single Stock Futures.

Yours sincerely,

N. PARAKH